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# Mobile Application Systems and Design
## Lab 5

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Q1. Dependencies Used

A. SQFlite: This is our sqlite plugin
I used it to perform SQL operations on local database

B. Path_provider: used to find correct system path.

C. Provider: used for state management

D. Intl: used for formatting timestamps

## Why SQLite is necessary for local storage

SQLite is necessary for this because it provides persistence and structure.

SQLite is relational database. It allows to store hundreds of posts search and sort the data

Q2 Handling database exceptions.

- Database not initialized: I used singleton pattern. Create a static database? database getter and getter that checks if database != null)
- Invalid/deleted data error: Using try-catch blocks
- Invalid or corrupted data: using model fromMap() class with validation logic

Q2 Explaining SQLite in Flutter.

Database vs table: database is the entire file stored on phone's disk
A table is a specific structure inside that file
thus columns, columns.

CRUD Operations:

1. Create: used db.insert() to add new post
2. Read: used db.query() to fetch all posts or single post by id
3. Update: used db.update() to modify existing post.
4. Delete: used db.delete() to remove logs from history.

Asynchronous Interaction:

Interaction is handled via future, async and await.

# Application Screenshots

Figure 1: Home Screen - List of Posts

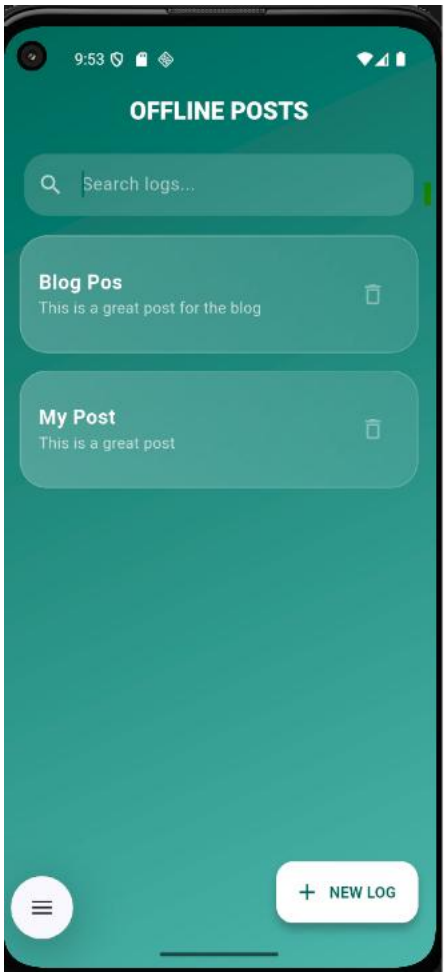

Figure 2: Add Post Screen

Figure 3: Post Details Screen

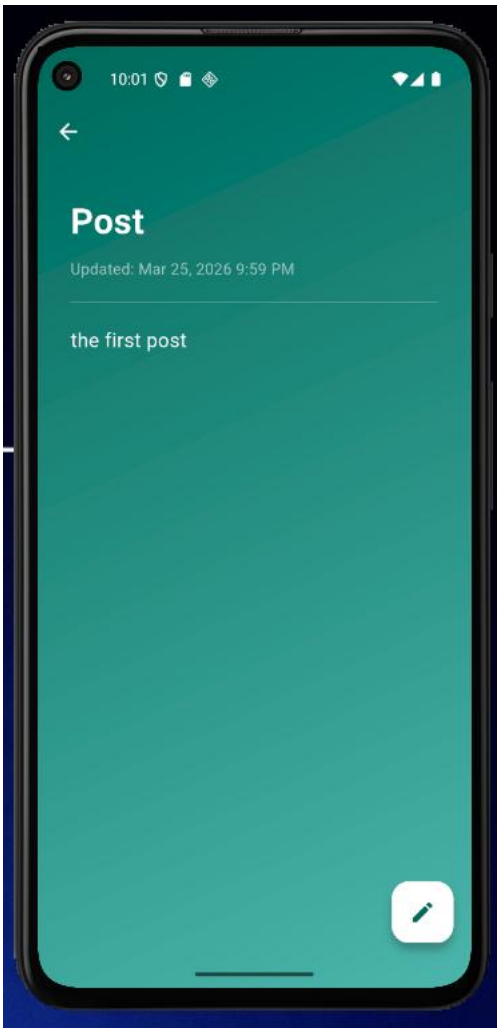

Figure 4: Edit Post Screen
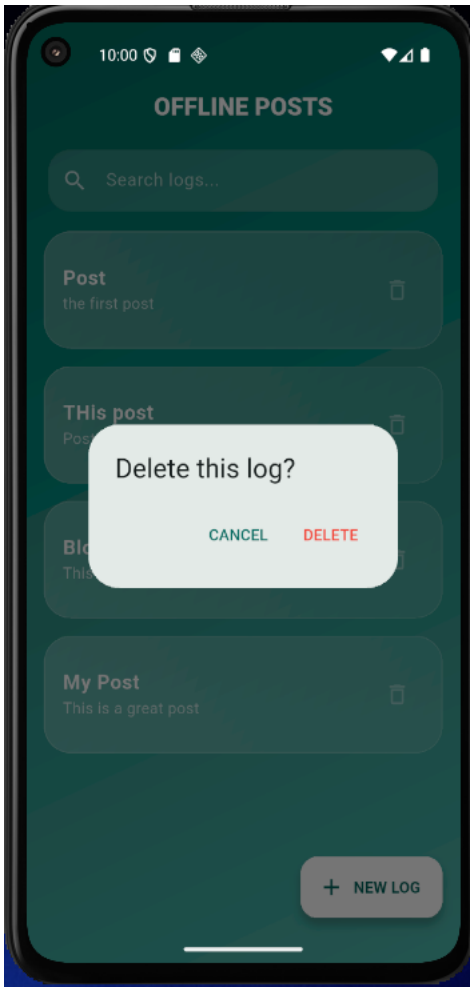
Figure 5: Delete Confirmation Dialog